ABB Linear Algebra 2026 Midterm Solutions

1. Find the 3×3 matrix A such that

$$\begin{bmatrix} 3 & 2 & 0 \\ 1 & 2 & 2 \\ 4 & 2 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution

Using the column view of multiplication we see that its first column is $\begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$, then this plus the second column is $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ so the second column is

$$\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}.$$

Getting the third column by subtracting $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ from $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ gives that

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 1 & 1 & 0 \\ 4 & -2 & -1 \end{bmatrix}.$$

2. Explain why if R is an invertible upper-triangular matrix, then R^{-1} is also upper triangular.

Solution

As R is invertible, using Gaussian Elimination, we can get from R to I by multiplying on the left by matrices that add copies of a row j to a row $i < j$, and by matrices that scale a row. All such matrices are upper triangluar, so their product, which is R^{-1}, is also upper triangular.

Alternatley, (avoiding Gaussian Elimination) let $R = \begin{bmatrix} r_1 \mid r_2 \mid \cdots \mid r_n \end{bmatrix}$ and $R^{-1} = [a_{ij}]$. Fixing j let i_j be the maximum value such that $a_{i_j j} \neq 0$. To show that R^{-1} is upper-triangular we show that $i_j \leq i$. By the column view of multiplication we have that where e_i is the ith column of I,

$$e_i = a_{1i}r_1 + a_{2i}r_2 + \cdots + a_{ni}r_n = a_{1i}r_1 + a_{2i}r_2 + \cdots + a_{i_j i}r_{i_j}.$$

As R is invertible the i_j entry of r_{i_j} is non-zero, and so the i_j entry of e_i is non-zero. Thus $i_j = i$, and so $i_j \leq i$, as needed.

3. Show that if a matrix A is upper-triangular and **unitary**, then it is diagonal.

Solution

That A is unitary means that it is square, and its columns are orthanormal, so $A^* = A^{-1}$ is its inverse. As it is upper-triangular, we have by the previous question that A^* is upper-triangular and so A is lower-triangular. Being upper and lower triangular, it is diagonal.

Alternately, you could argue that as the columns are orthogonal, they are unchanged by removing the projections onto earlier columns, and so show by induction that each column is 0 above the diagonal.

4. Let $\|\cdot\|$ be a vector norm. Define a matrix norm $\|\cdot\|_M$ from the vector norm, and show that it is **submultiplicative**.

Solution

We define our matrix norm by

$$\|A\|_M = \max_{v \in \mathbb{C}^m \setminus \{0\}} \frac{\|Av\|}{\|v\|}.$$

To see that $\|AB\|_M \leq \|A\|_M \|B\|_M$ and get that $\|\cdot\|_M$ is submultiplicative we compute as follows:

$$\begin{aligned} \|AB\|_M &= \max_{v \in \mathbb{C}^m} \frac{\|ABv\|}{\|v\|} = \max_v \frac{\|ABv\|}{\|Bv\|} \frac{\|Bv\|}{\|v\|} \\ &\leq \max_v \frac{\|ABv\|}{\|Bv\|} \cdot \max_v \frac{\|Bv\|}{\|v\|} \\ &= \|A\|_M \|B\|_M. \end{aligned}$$

The inequality in the second line is because the v maximising $\frac{\|ABv\|}{\|Bv\|} \frac{\|Bv\|}{\|v\|}$ need not maximise either of these factors individually.

5. Show for any vector $v \in \mathbb{C}^m$ that

$$\|v\|_2 \leq \sqrt{m} \|v\|_\infty.$$

($\|v\|_\infty$ is a **bold term**.)

Solution

Where $v = (v_1, \ldots, v_m)$ we have $\|v\|_\infty = \max_{i=1}^m |v_i|$ and

$$\|v\|_2 = \sqrt{\sum_{i=1}^m v_i^2} \leq \sqrt{m \cdot (\max_{i=1}^m v_i)^2} = \sqrt{m} \cdot \max_{i=1}^m |v_i| = \sqrt{m} \cdot \|v\|_\infty.$$

6. Define a Singular Value Decomposition of a matrix, and state the Theorem about their existence.

Solution

Every matrix A has an SVD $A = U\Sigma V^*$ where U and V are unitary matrices, and Σ is a (not necessarily square) diagonal real matrix. If Σ is square and its diagonal entries are distinct non-zero numbers in decreasing order, then the columns of U and V are unique upto scalar factors ± 1.

7. Show that if P is a **projector**, then $I - P$ is too.

Solution

A matrix $\boxed{A \text{ is a projector if } A^2 = A}$. Assuming that P is a projector, we get

$$(I - P)^2 = I^2 - 2P + P^2 = I - 2P + P = (I - P)$$

showing that $(I - P)$ is a projector.

8. Let E be the matrix such that $Ex = (x + Fx)/2$ where F is the F is matrix such that

$$F \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} x_m \\ \vdots \\ x_1 \end{bmatrix}.$$

Describe E and decide if it is an **orthogonal projector**, an oblique projector, or not a projector at all.

Solution

As F is the back-diagonal matrix

$$F = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 0 \\ & & \vdots & & \\ 0 & 1 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \end{bmatrix}$$

with $F^2 = I$ and $E = \frac{1}{2}(I + F)$ is the matrix with an X of entries $\frac{1}{2}$; if m is odd, then the center entry is 1. From this we observe that $E^* = E$. Observing that $E - I = \frac{1}{2}(F - I)$, we get

$$E^2 = (\frac{1}{2}(I + F))^2 = \frac{(I + F)^2}{4} = \frac{I + F + F + F^2}{4} = \frac{1}{2}(I + F) = E \quad (1)$$

$$E^*(E - I) = \frac{(I + F)(F - I)}{4} = \frac{F + F^2 - I - F}{4} = \frac{F + I - I - F}{4} = 0, \quad (2)$$

showing that E is an orthogonal projection.

9. Give a reduced QR decomposition and a full QR decomposition of the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Solution

The colums are already orthogonal, so we just have to normalise them to get the reduced QR decomposition

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} =: \hat{Q}\hat{R}.$$

For the full QR decomposition we must add an orthogonal column to $\hat{Q}$

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} =: QR.$$

10. Write out any of the algorithms for finding a QR factorisation of a matrix A, and compute the running time.

Solution

See text.